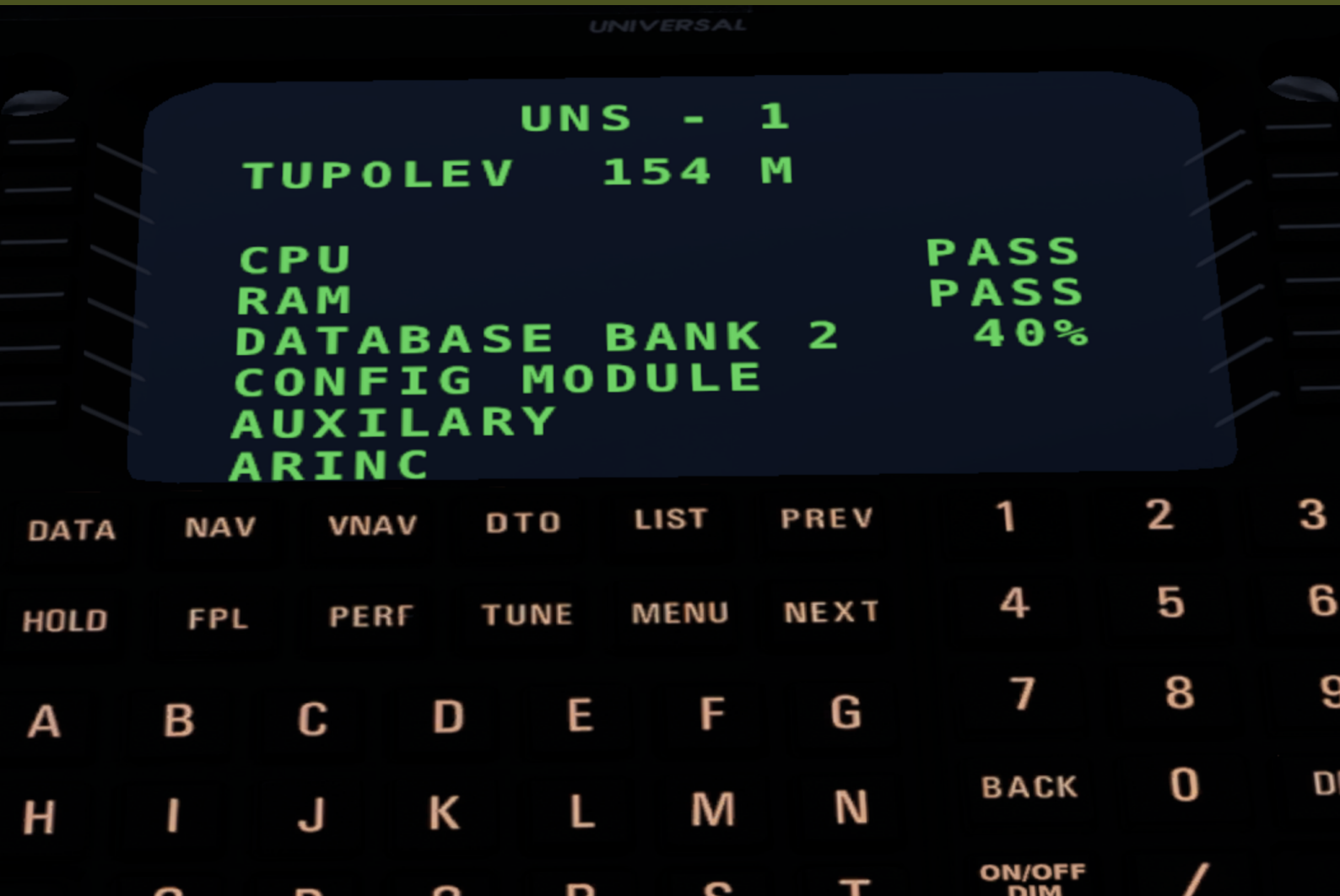



# UNS-1 FMS XP12 OPERATIONS MANUAL

**created for TU-154M V3 · X-PLANE 12**

**Freeware addition for Felis Community continuation mod**



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# INTRODUCTION

This manual provides details of how to use the Universal UNS-1 Flight Management System (FMS) which is integrated into the Kontur cockpit of the Tu-154M v3 mod for X-Plane 12.

The UNS-1 looks and is operated like a Universal UNS-1: its screen layout, colours, cursor boxes, prompts and keys follow the real unit. Behind the screen it uses X-Plane 12's own FMS, so there is one flight plan for the whole aircraft: the UNS-1 builds it, the two Kontur displays draw it and the autopilot (ABSU) can follow it. Runways, SIDs, STARs and approaches come from X-Plane's navigation database.




There are two UNS-1 units on the centre panel: the captain's (left) and the right one. They share the flight plan: a change made on one unit (ORIGIN, DEST, a waypoint, a direct-to) shows on the other one at once. Everything else is independent: each unit shows its own page (one pilot can be on FPL while the other is on NAV or a DATA page), and each has its own power (ON/OFF DIM key), self test, INIT page, DIM page and brightness. A 2D panel of the captain's unit can also be opened (see POWER AND BRIGHTNESS).

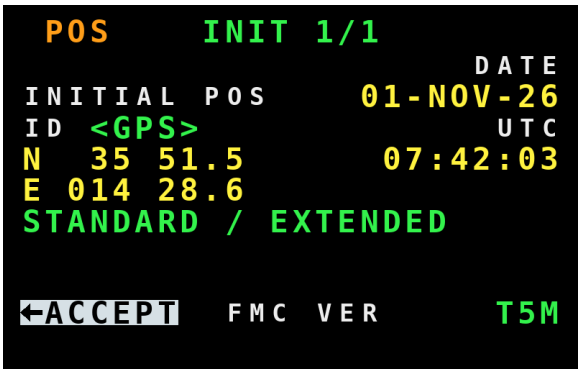
## Page layout

Most pages are presented with this common structure:

- **Page title** – shows the name of the current page on the top line of the page.
- **Page number** – shows the current page number and the total number of available pages on the right of the title line (e.g. `1/2`).
- **Content** – page content is shown on the following lines. If a line has an arrow that points to the adjacent key (`←FIX`, `RTN→`), it is an action that can be triggered by pressing that key.
- **Cursor** – shown as highlighted text (dark text on a light box). It marks a selected item, an entry being typed or something that needs to be validated.
- **RTN** – pressing the adjacent key returns the UNS-1 to the previously displayed page.
- **Colours** – labels and prompts are white, values green, the active waypoint magenta, messages and `MOD` amber. **Blue** marks what is not available or not operative: a value that is not known yet (`----`, `--:--`), a display-only page (TUNE) or a function that the Tu-154M UNS-1 does not have (`AWY→` on LIST).

- **Framed H** – on the FPL, DTO and PATH VNAV pages a framed H marks the altitude column.
- **Entry fields** – a white field with black dashes is waiting for an entry (ORIGIN, DEST, HOLD FIX, the # field on LIST). What is typed appears in a white field, either at the field itself or at the bottom of the screen, ready for the key of the field it is meant for.
- **MOD and ENTER** – MOD in amber in the title shows a change that is not yet active. ←ENTER then appears next to LSK 6: press it, or the ENTER key, to activate the change.
- **Messages** – short messages appear in amber on the title line for about four seconds (see MESSAGES (MSG) PAGE).

## Examples




### INIT page

Page number 1/1 shows that no other pages are available.

The cursor is highlighting the ←ACCEPT text, indicating that it needs to be validated by pressing LSK 6 or the ENTER key.




### FPL page

The page number 1/2 shows that page 1 is currently displayed and one other page is available, accessible by pressing the NEXT key.

The waypoints are numbered; the white box on line 5 is a route discontinuity.

## General usage

The FMS is composed of a screen and keys. The keys located immediately around the screen are called select keys and are used according to the page content. The six select keys on the left are the Left Select Keys (LSK) 1 to 6 and the ones on the right are the Right Select Keys (RSK) 1 to 6. They are referred to in this way throughout this manual.

Below the screen are the function keys (two rows), the alphanumeric keys and the numeric keys used for data entry. The role of each function key is explained in depth later in this manual.




The UNS-1 keys (2D panel shown; the 3D units are identical).

Some keys have the same function, whichever page is displayed:

- When multiple pages are available, shown in the title line with the page number:
  - `NEXT` shows the next page
  - `PREV` shows the previous page
- `MENU` always opens FPL MENU, the start page with origin, destination and the main menu.
- `LIST` opens the LIST page: the nearest navaids, airports and fixes, or the flight plan's points, to pick by number instead of typing a name (see LIST PAGE).
- When user validation is required, the `ENTER` key is used. It is also offered as `←ENTER` next to LSK 6.
- `BACK` erases the last character typed (or clears the scratchpad message), `DEL` deletes.
- The `ON/OFF DIM` key can be used with any page to adjust the screen brightness or turn the unit on/off (see POWER AND BRIGHTNESS).
- The UNS-1 keypad has no decimal point key: enter values without it.

# QUICK START GUIDE

This section describes the way this FMS should be used to plan and fly a route, from departure to destination. It shows a typical sequence for the use of the different FMS pages that are explained in further detail in the following sections of this manual.

For a full tutorial flight, please refer to the FLYING WITH THE UNS-1 section of the manual.

### On the ground, prior to take-off

1. Both UNS-1 units are off when the flight is loaded, also with the engines running. Switch on the UNS power (GPS toggle, the last toggle of the upper overhead row, with the electrical buses powered; already on if the flight was loaded with engines running), then press `ON/OFF DIM` on each unit. Turn a unit off and back on using `ON/OFF DIM` and `OFF→` if you wish to reset it.
2. Check the initial aircraft position on the INIT page. The GPS position, UTC date and time are set automatically and just need to be cross-checked and confirmed by pressing LSK 6 `←ACCEPT` or `ENTER`.
3. Create the flight plan:
   - On FPL MENU, input the departure airport (LSK 1) and the destination airport (RSK 1).
   - Select the departure runway and SID for departure (FPL MENU, LSK 3 `←DEP/ARR SID/STAR`).
   - Manually input the waypoints on the FPL page, or enter airways on X-Plane's route page (see Using airways).
   - Select the arrival runway and STAR for arrival (if available).
4. Set the cruise altitude on the VNAV CLB page (after the PATH VNAV pages: CRZ ALT).

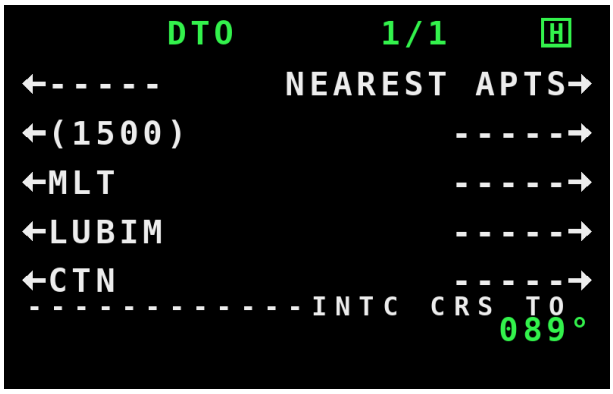
When all this is done, the FMS is capable of managing lateral navigation. The flight plan is shown on the Kontur displays. For the autopilot to follow it, use the ABSU navigation (НВУ) mode with the UNS–CHC switch (between the Kontur displays) at UNS; please refer to the Tu-154M v3 documentation for the ABSU.

### During the flight

1. The FPL page shows the course and distance to each waypoint of the flight plan.
2. The NAV page can be displayed to check the aircraft's progress along the programmed route, especially the times over the waypoints, the cross-track error and the wind.
3. The DIRECT TO (DTO) page can be used if Air Traffic Control clears you to fly direct to a waypoint.
4. The VNAV pages can be used for the cruise level, speeds and the descent.

# DIRECT TO (DTO) PAGE

Pressing the **DTO** function key brings up the 'direct to' page. This page is used to fly directly to any waypoint in the flight plan.



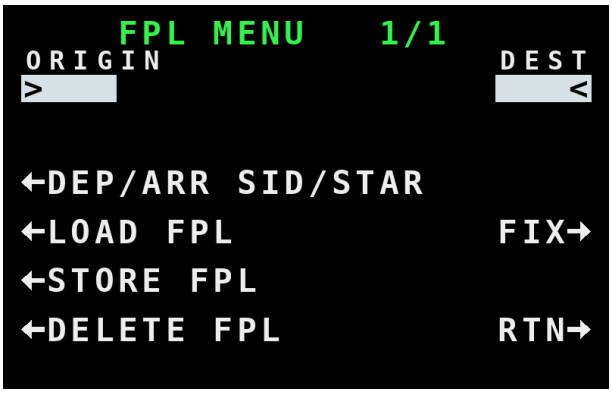

The DTO page lists the next waypoints of the flight plan next to LSK 2 to 5, by name only, with their altitudes on the right (framed ). To fly direct to one of them, press its LSK. The UNS-1 makes it the active waypoint (magenta), validates the change and shows the FPL page with the message **DIRECT TO …**. The Kontur and the autopilot then fly from the present position to that waypoint.

To fly direct to a point that is not in the flight plan, type its name, press LSK 1 and then **ENTER**. RSK 1 **NEAREST APTS→** lists the nearest airports.

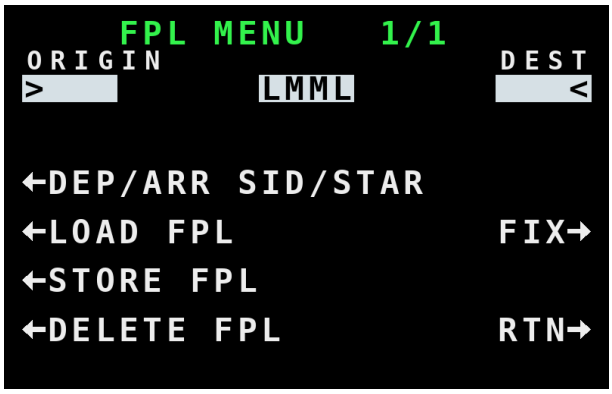
A direct-to can also be made from the FPL page: select the waypoint and press RSK 2 **DTO→** (see Waypoint insertion and deletion). If the direct-to is refused, the message is **DTO NOT ACCEPTED**.

# FLIGHT PLAN (FPL) PAGE

The flight plan is started on FPL MENU, which opens with the **MENU** key and is the first page after the INIT page. Enter the departure airport first: type its ICAO code (e.g. LMML) and press LSK 1 (ORIGIN). Then type the destination airport and press RSK 1 (DEST). As in a real FMS, entering a new departure airport starts a new flight plan.




FPL MENU before any entry: ORIGIN and DEST are empty fields.




Typing the departure airport (4 letters).

Each entry takes about a second: the UNS-1 enters the airport into X-Plane's flight plan and validates it. WAIT is shown meanwhile; then the airport appears under ORIGIN or DEST with the message **ORIGIN LMML**.

| Key on FPL MENU | Action |
|---|---|
| LSK 1 ORIGIN / RSK 1 DEST | Enter the typed airport as departure / destination. Empty, the fields are white boxes: `>` at ORIGIN, `<` at DEST. |
| LSK 3 **←DEP/ARR SID/STAR** | Runways, SIDs, STARs and transitions. |
| LSK 4 **←LOAD FPL** | The list of stored flight plans (STORED FPL), to load one (see Other flight plan functions). |
| LSK 5 **←STORE FPL** | Store the flight plan as a file (see Other flight plan functions). |
| LSK 6 **←DELETE FPL** | Delete the whole flight plan. First press: **←CONFIRM DELETE** in amber; second press: deleted (**FPL DELETED**). Any other key cancels. |
| RSK 4 **FIX→** | X-Plane's FIX page (reference circles and radials around a fix on the map). |
| RSK 6 **RTN→** | The FPL page. |
| **BACK** / **DEL** | Erase the last letter / the whole entry. |

The FPL page ( **FPL** ) shows the flight plan as a list of points, numbered **1>**, **2>** …, with the speed / altitude constraints on the right under the framed **1**. The active waypoint is magenta. Above each point is the course and distance of the leg to it. A route discontinuity is a white field with ROUTE DISCONTINUITY above it. Use **NEXT** and **PREV** to page through a long flight plan. Times to the waypoints are on the NAV and PERF pages. LSK 6 **←AWY** opens the AIRWAYS page (see Using airways).




The FPL page.

## Duplicate waypoint names

Several waypoints with the same name may exist. If this is the case, the SELECT WPT page appears when the name is entered, with one line per waypoint and its position. Select the right one with its LSK.

## Using airways

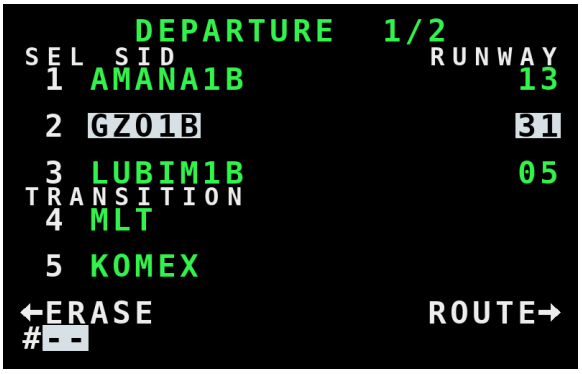
Airways (AWY) are entered on the AIRWAYS page (X-Plane's FPLN page), which has a VIA (airway) and a TO (waypoint) column:

1. On the FPL page press LSK 6 `←AWY`. The AIRWAYS page opens. (It can also be reached from STORED FPL, LSK 5 `←AWY`.)
2. Type the airway name (e.g. `UM601`) and press the left key of the first empty VIA line (LSK 4 on a new route).
3. Type the waypoint where you leave the airway and press the right key of the same line (RSK 4, TO).
4. Press `ENTER`. The waypoints of the airway are added to the flight plan; check them on the FPL page.

The UNS-1 LIST page cannot show airways: `AWY→` is blue.

## Managing runways, SIDs/STARs and transitions

The runways and procedures of the departure and destination airports are selected through FPL MENU, LSK 3 `←DEP/ARR SID/STAR`. The DEPARTURE and ARRIVAL pages come from X-Plane's FMS and are shown the UNS way: the SIDs / STARs / transitions are numbered on the left (`SEL SID`), the runways / approaches are on the right, and what is selected is shown in a cursor box. Select an item with its select key, or type its number (it appears in the `#` field) and press `ENTER`.




**DEPARTURE page**

SID `GZ01B` and runway `31` are selected (cursor boxes).

Typing `3` and `ENTER` selects `LUBIM1B`, the same as LSK 3.

A number works for the items shown on the current page; `NEXT` / `PREV` show the other pages.

### Standard Instrument Departure (SID)

1. On the DEP/ARR page press LSK 1 `←DEP` for the departure airport.

2. Select the departure runway, then the SID, then the transition if one is offered (select key, or number + `ENTER`). Use `NEXT` / `PREV` if the list has more pages.
3. The title shows `MOD` and `←ENTER` appears next to LSK 6. Press it or `ENTER`.
4. Press `FPL`: the SID waypoints are now in the flight plan.

### Standard Terminal Arrival Route (STAR)

1. On the DEP/ARR page press RSK `ARR→` for the destination airport.
2. Select the approach / runway, then the STAR and its transition.
3. Press `ENTER` (or LSK 6 `←ENTER`) and check the result on the FPL page. Close any discontinuity that appears (see below).

## Waypoint insertion and deletion

### Inserting a waypoint

1. On the FPL page, type the waypoint name (e.g. `MLT`).
2. Press the LSK of the waypoint it should come *before*. To add at the end, press the LSK of the first empty line or of the discontinuity.
3. If the SELECT WPT page appears, choose the right waypoint.
4. Press `ENTER`.

### Deleting a waypoint

With the scratchpad empty, press the LSK next to a waypoint (lines 1 to 5). The waypoint is highlighted by the cursor and the UNS options appear on the right:




Waypoint 3 selected: DEL→, DTO→, HOLD→, RTN→.




After the first DEL→: CONFIRM→ to delete.

| Key | Action |
|---|---|
| RSK 1 `DEL→` (or `DEL`) | First press: `CONFIRM→` appears in amber. Second press: the waypoint is deleted and the change validated (`MLT DELETED`). |
| RSK 2 `DTO→` | Fly direct to this waypoint (`DIRECT TO MLT`). |
| RSK 3 `HOLD→` | A holding pattern at this waypoint (see HOLDING). |
| RSK 6 `RTN→`, `BACK` or the same LSK | Cancel the selection. |
| Another LSK | Select that waypoint instead. |

A route discontinuity (white box with ROUTE DISCONTINUITY above it) is removed the same way: select it, then `DEL→` and `CONFIRM→`. The UNS-1 joins it to the next waypoint (`DISCONTINUITY CLOSED`). If a waypoint cannot be deleted (for example some procedure points), the message is `CANNOT DELETE …`.

## Flight plan summary page

*There is no separate summary page in the Tu-154M UNS-1. Distances and times along the route are shown on the NAV and PERF pages.*

## Other flight plan functions

### Flight plan deletion

On FPL MENU press LSK 6 `←DELETE FPL` twice (the first press shows `←CONFIRM DELETE` in amber). Entering a new departure airport also starts a new flight plan. The flight plan is also erased completely when no UNS-1 unit is on any more: both switched off with `ON/OFF DIM` / `OFF→`, or the UNS power is cut. The first unit switched on again starts with an empty flight plan.

### Flight plan import (LOAD FPL)

Routes saved as X-Plane .fms files in the folder *X-Plane 12/Output/FMS plans* (by SimBrief, Navigraph, Little Navmap, or STORE FPL below) can be loaded: on FPL MENU press LSK 4 `←LOAD FPL`. The UNS-1 goes through X-Plane's FPLN and ROUTE MENU pages by itself and shows the list of stored flight plans (STORED FPL, X-Plane's company route list). Select the route and press `ENTER`. Check the result on the FPL page. If the list cannot be reached, the message is `CO ROUTE LIST NOT FOUND`.

### Flight plan export (STORE FPL)

On FPL MENU press LSK 5 `←STORE FPL`. The flight plan is written as a standard X-Plane flight plan file into *X-Plane 12/Output/FMS plans*, named after the route (e.g. *LMML-LICC.fms*), with the AIRAC cycle of the navigation data in use; the message is `SAVED LMML-LICC`. Saving the same route again replaces the file. The route is saved as its waypoints (and the departure runway): after loading it, select the SID, STAR and approach again on DEP/ARR.

## Updating the Nav database

The UNS-1 uses X-Plane 12's navigation database. To update it, update X-Plane's navigation data (for example with a Navigraph subscription, installed into *X-Plane 12/Custom Data*). When X-Plane reports that its data is out of date, the INIT page shows `"NAV DATABASE EXPIRED"`; `←ACCEPT` also clears X-Plane's own NAV DATA OUT OF DATE message, as CLR does on the Boeing CDU.

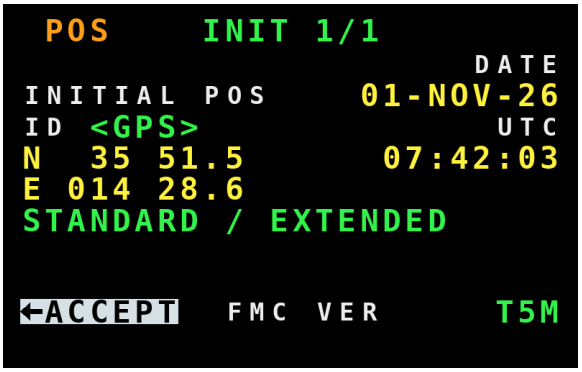
# FUEL MANAGEMENT (FUEL) PAGE

*Fuel management is not part of the Tu-154M UNS-1: there is no FUEL key on its keypad (the HOLD key is in that place). Fuel is managed with the Tu-154M's own EFB, fuel system and gauges.*

## Fuel menu (FUEL OPTIONS) page

*This function is not available in the Tu-154M UNS-1.*

# INIT PAGE




After the self test of a unit, the INIT page (POS INIT) is shown on that unit.

It shows the aircraft's present position from GPS, the UTC date and the UTC time from the simulator, the database type and the FMC version.

Check the data and press LSK 6 `←ACCEPT` or `ENTER`. Until then the other keys of that unit do nothing. ACCEPT also clears X-Plane's NAV DATA OUT OF DATE message on that unit. After acceptance the unit shows FPL MENU.

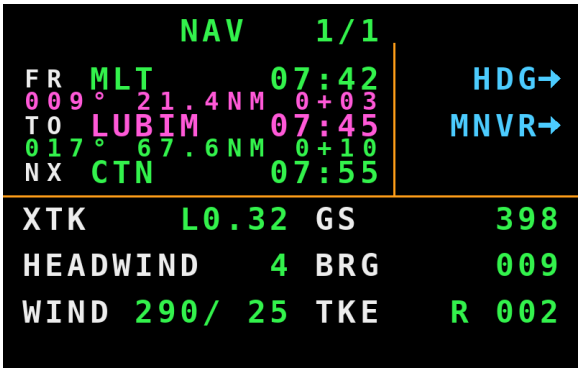
# MESSAGES (MSG) PAGE

The Tu-154M UNS-1 has no MSG key and no message page: messages appear in amber on the title line for about four seconds, then the page title comes back.

| Message | Meaning |
|---|---|
| `ORIGIN LMML` / `DEST LIRF` | Airport entered and validated. |
| `ORIGIN NOT FOUND` | The UNS-1 could not reach X-Plane's route page; enter the airport on the route page (FPLN). |
| `ORIGIN NOT ACCEPTED` | The airport was refused (not in the database, or mistyped). |
| `TYPE ICAO, THEN KEY` | On FPL MENU: type the airport first, then press LSK 1 or RSK 1. |
| `MLT DELETED` / `FPL DELETED` | Waypoint / whole flight plan deleted. |
| `CANNOT DELETE MLT` | This waypoint cannot be deleted. |
| `DISCONTINUITY CLOSED` / `CANNOT CLOSE GAP` | Gap removed / no waypoint after the gap to join it to. |
| `DIRECT TO MLT` / `DTO NOT ACCEPTED` | Direct-to done / refused. |
| `MLT NOT ON DTO PAGE` | The waypoint is not in the list of the DTO page. |
| `SAVED LMML-LICC` / `NO ROUTE TO SAVE` / `SAVE FAILED` | STORE FPL: done / no route / the file could not be written. |
| `NOT IN DATABASE` | The identifier is not in X-Plane's navigation database. |
| `INVALID ENTRY` / `NOT ON THIS PAGE` | The typed value is not accepted / the number is not on the page shown. |
| `V/S 100 TO 6000` | PATH VNAV: target vertical speed out of range. |
| `HOLD NOT AVAILABLE` / `NO POINT AFTER LUBIM` | HOLD: X-Plane offers no hold here / EXIT HOLD has no point to continue to. |
| `AIRWAYS NOT AVAILABLE` | LIST: airways cannot be listed. |

| Message | Meaning |
|---|---|
| `CO ROUTE LIST NOT FOUND` | LOAD FPL: X-Plane's list of stored flight plans could not be opened. |
| `HDG NOT AVAILABLE` / `KEY INOPERATIVE` | NAV: `HDG→` / `MNVR→` are not available (blue). |
| `MOD` (title) | A change is waiting for `ENTER`. |

# NAVIGATION (NAV) PAGE




Press NAV. The page shows the progress along the route:

- **FR / TO / NX** – the waypoint just passed, the active waypoint (magenta) and the next one, each with a time (UTC): the time over FR, and the estimated times of arrival at TO and NX.
- **Magenta line** (under FR) – the active leg: its course, the distance and the time (h+mm) to go to TO.
- **Green line** (under TO) – the next leg: its course, length and time.
- **XTK / GS** – cross-track distance (L or R) and ground speed.
- **HEADWIND (TAILWIND) / BRG** – wind component and bearing to the active waypoint.
- **WIND / TKE** – wind direction / speed and track angle error (L or R).

The time over FR is noted when the aircraft passes a waypoint; until the first one is passed it is --:--. A value that is not available is blue. LSK 2 (the TO line) opens the DTO page.

## Heading (HDG) page

*Not available: HDG→ (RSK 1) is blue. Heading is selected on the Tu-154M's own autopilot panel.*

## Maneuver (MNVR) page

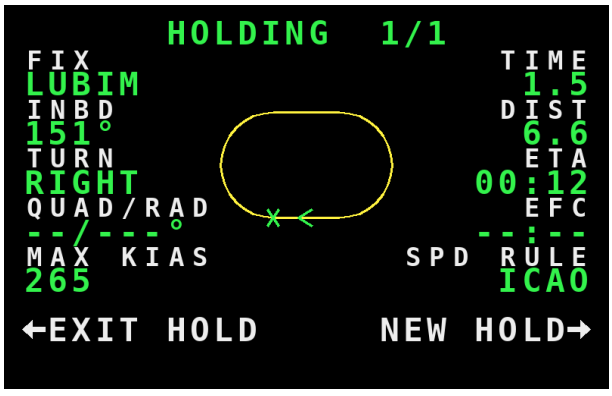
*Not available: MNVR→ (RSK 2) is blue. Holding patterns have their own HOLD key (see below).*

## Holding (HOLD)

The HOLD key opens the HOLD FIX page: the points of the flight plan, numbered, and the HOLD FIX field. Choose the holding fix with the LSK beside a point, or type its number or any identifier and press RSK 1 or ENTER. RSK 6 PPOS→ holds at the present position, when X-Plane offers it. A hold can also be started from the FPL page: select the point and press RSK 3 HOLD→.




HOLD FIX: the points of the flight plan.




HOLDING: the hold at LUBIM.

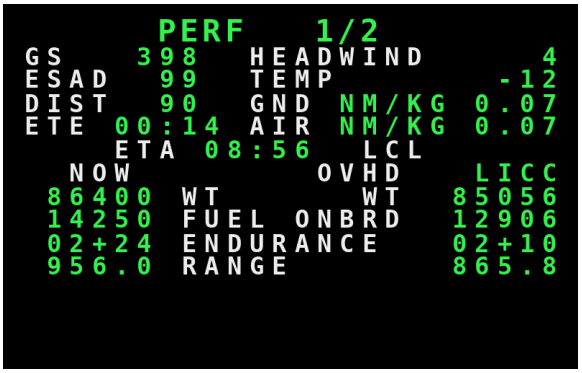
The HOLDING page then shows the hold with MOD and ←ENTER: check it and press ENTER to activate it. The values are X-Plane's; each key changes the field next to it:

| Key | Field |
|---|---|
| LSK 1 FIX | Holding fix. A typed identifier + LSK 1 makes a new hold at that fix. |

| Key | Field |
|---|---|
| LSK 2 INBD | Inbound course (type e.g. `160`). |
| LSK 3 TURN | RIGHT / LEFT: press to change, or type `L` / `R` first. |
| LSK 4 QUAD/RAD, LSK 5 MAX KIAS | Quadrant / radial and maximum holding speed. |
| RSK 1 TIME, RSK 2 DIST | Leg time (minutes) and leg distance (NM). |
| RSK 3 ETA (blue), RSK 4 EFC | Time at the fix (display only) and expected further clearance time. |
| RSK 5 SPD RULE | Holding speed rule (FAA / ICAO). |
| LSK 6 `←EXIT HOLD` | Leave the hold: the UNS-1 flies direct to the flight plan point after the holding fix. Shown when the aircraft is at the hold. |
| RSK 6 `NEW HOLD→` | Another hold (back to HOLD FIX). |

The racetrack in the middle shows the inbound leg with an arrow to the fix (cross); it is drawn the other way round for left turns. Holds that come with a procedure (for example the missed approach) stay in the flight plan and are flown as the procedure says; the `HOLD` key always starts with the HOLD FIX list, and the hold made on the UNS-1 opens with `HOLD` until it is exited. When the flight plan has several holds, X-Plane's list of holds is shown first: select one with its LSK.

## PERFORMANCE (PERF) PAGE




Press `PERF` (page 1/2). All values are computed from the aircraft and the flight plan:

- **GS / HEADWIND (TAILWIND)** – ground speed and wind component.
- **ESAD / TEMP** – equivalent still air distance and outside air temperature.
- **DIST / ETE / ETA** – distance and time to the destination, arrival time (local).
- **GND / AIR NM/KG** – ground and air nautical miles per kg of fuel.
- **NOW / OVHD** – weight, fuel on board, endurance and range now and over the destination.

`NEXT` shows page 2/2: X-Plane's PROGRESS page in UNS style.

### Performance menu (PERF MENU) page

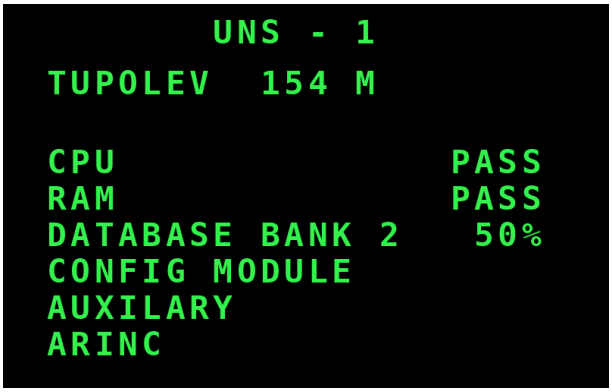
*This function is not available in the Tu-154M UNS-1.*

# POWER AND BRIGHTNESS

The `ON/OFF DIM` function key gives access to the unit's power and brightness control. It can be used from any page. Each of the two UNS-1 units has its own key and works independently.

Both units are off when a flight is loaded, also with the engines running. A unit comes on only when the UNS power is on (GPS toggle, the last toggle of the upper overhead row; already on if the flight was loaded with the engines running) *and* its `ON/OFF DIM` key is pressed.

If the unit is off, pressing `ON/OFF DIM` turns it on. It always starts from the beginning: a 17 s self test, then the INIT page, then the FPL MENU page (with the ORIGIN and DEST already entered, if the other unit is on).




Self test in progress.




The DIM page.

If the unit is on, pressing `ON/OFF DIM` brings up its DIM page:

| Key | Action |
|---|---|
| LSK 1 `←BRIGHT` / LSK 2 `←DIM` | Screen brightness of this unit in 10 % steps (starts at 20 %). Each unit has its own setting. |
| RSK 1 `OFF→` | Turns this unit off: its screen goes dark and its keys do nothing. |
| RSK 6 `RTN→`, `BACK` or `ON/OFF DIM` | Closes the DIM page. |

**Flight plan.** While at least one unit is on, the flight plan is kept: one unit can be turned off and on (it recycles) while the other carries on. When no unit is on any more - both turned off, or the UNS power cut - the flight plan is erased. After a power cut both units stay off until their `ON/OFF DIM` keys are pressed again.

**Panel lighting.** The UNS brightness knob on the centre panel dims the night lighting of the UNS panels (the lit key labels), not the screens. Like the other cockpit lighting knobs it keeps its last position: it is saved with the cockpit state of the livery and set back when the aircraft is loaded.

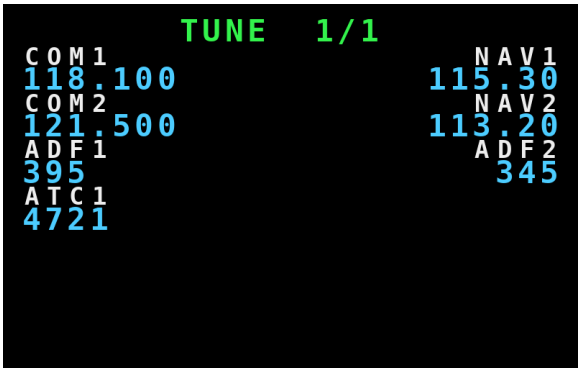
## 2D panel

The 2D panel is the captain's unit, with the same screen, keys and power; it is always shown at full brightness, whatever the DIM page setting. Open it by clicking the captain's UNS screen in the cockpit or with Aircraft > UNS > UNS-1 2D panel. Move it by dragging its frame (anywhere except a key or the screen); close it with the red dot in the top-right corner.

## UNS-1 or X-Plane FMC

The Aircraft > UNS menu chooses which FMS is on the cockpit screens. **UNS-1 FMS** (ticked by default) shows the UNS-1. **X-Plane FMC (default)** brings back X-Plane's own FMC: its Boeing-style CDU screen and popup, with the keys working as on that CDU. Both use the same X-Plane flight plan. Tick UNS-1 FMS to return to the UNS-1. **UNS-1 2D panel** opens or closes the 2D panel.

# TUNE PAGE

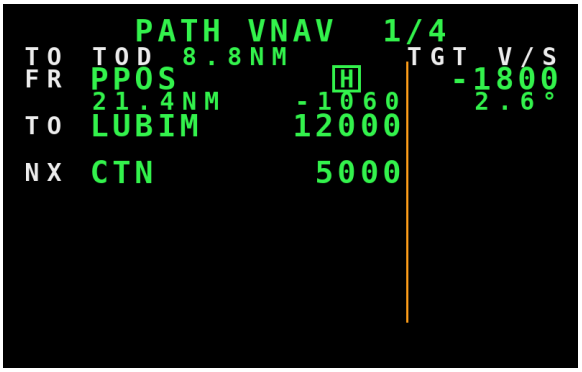



Press **TUNE**. The page shows the frequencies of COM1/2, NAV1/2, ADF1/2 and the transponder code (ATC1).

As in a Tu-154M with its own radio panels, the radios are tuned on those panels, not on the UNS-1: the TUNE page is display only, so its values are blue.

# VERTICAL NAVIGATION (PATH VNAV) PAGE

The **VNAV** key opens the UNS-1 PATH VNAV page. It runs over as many pages as the flight plan needs (page 1: TO and four NX points; each following page: five more points, to the end of the plan). After them come X-Plane's own CLB, CRZ and DES pages (in UNS style). The page number counts them all (with a short plan: PATH VNAV 1/4, VNAV CLB 2/4, CRZ 3/4, DES 4/4). **NEXT** / **PREV** move through them; **VNAV** always opens page 1.




**PATH VNAV (page 1)**

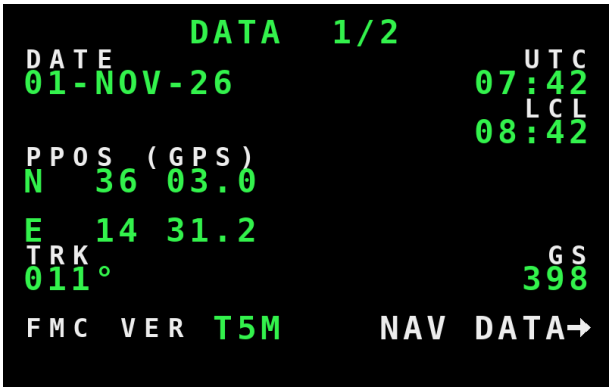
- **TO** – the next waypoint with an altitude in the flight plan; **NX** – the following ones (only points with an altitude are listed). Their altitudes are under the framed **H**.
- **FR PPOS** – the distance to TO and the vertical speed needed now to be at its altitude there.
- **TGT V/S** – the planned descent rate: press RSK 1, type it (e.g. **1800**; it is shown as **-1800**) and press **ENTER**. The angle of that descent at the present ground speed is shown under it, and **TO TOD** gives the distance to the top of descent until it is reached.



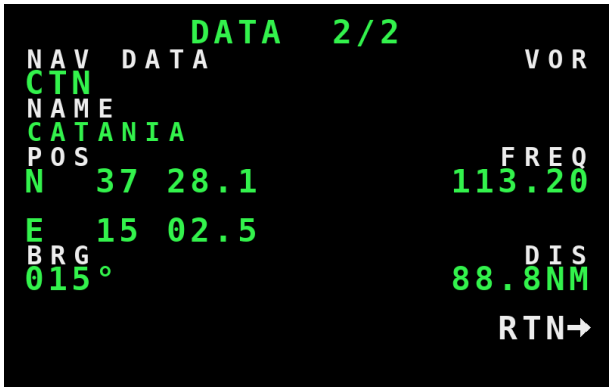

**VNAV CLB**

X-Plane's climb page. The cruise altitude is entered here: type it (e.g. **330** or **33000**) and press the LSK of CRZ ALT, then **ENTER** if **MOD** is shown. The CRZ and DES pages that follow work the same way.

## DATA PAGE



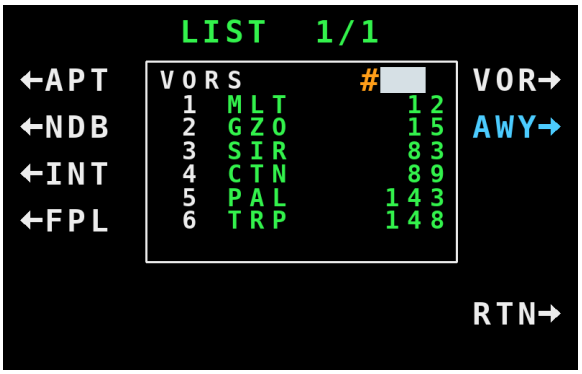

DATA 1/2.




DATA 2/2: NAV DATA for CTN.

Press **DATA**. Page 1/2 shows the date, UTC and local time, the aircraft position (GPS), track, ground speed and the FMC version. **NEXT** or RSK 6 **NAV DATA→** opens page 2/2, NAV DATA: type an identifier and press LSK 1 or **ENTER** to see the nearest navaid, fix or airport of that name in X-Plane's database: its type, name, position, frequency and the bearing / distance from the aircraft. If other points have the same name, the page says how many.

## LIST PAGE




Press **LIST**. The nearest VORs are listed first, numbered and with their distance (NM), from X-Plane's navigation database. The keys change the list:

- LSK 1 **←APT** airports, LSK 2 **←NDB** NDBs, LSK 3 **←INT** intersections, LSK 4 **←FPL** the points of the flight plan, RSK 1 **VOR→** VORs.
- **NEXT** / **PREV** – more of the list (up to 10 pages).

Type the number of an item (it appears in the **#** field) and press **ENTER**. The identifier goes back to the page LIST was opened from: on FPL MENU into the ORIGIN / DEST entry, on DATA it is shown on NAV DATA, on the FPL, DTO and HOLD FIX pages it is ready for the LSK where it should go; from the other pages the FPL page opens with it. RSK 6 **RTN→** goes back without a choice. The first time a list is opened the UNS-1 searches the database for a few seconds (**SEARCHING**).

# FLYING WITH THE UNS-1

This tutorial flight covers a short flight with the Tu-154M v3 in the Kontur cockpit. It covers the procedures required to set up and operate the UNS-1, but not the operation of the aircraft itself; for that, please see the Tu-154M v3 documentation.

In this tutorial flight we depart from Malta International Airport (LMML) and fly north-east across the sea to Catania Fontanarossa (LICC) in Sicily. The runways and procedures used depend on the weather and on your navigation data; choose the ones in use on the day.

| Departure | Destination | Route |
|---|---|---|
| LMML Malta | LICC Catania | SID from LMML – en-route waypoints of your choice – STAR and approach for LICC |

## Pre-flight

1. With the aircraft powered up, switch on the UNS power: the GPS toggle (last toggle of the upper overhead row). If the flight was loaded with the engines running, it is already on.
2. Press `ON/OFF DIM` on the captain's unit, and on the right unit if you want to use it. Each unit runs its self test.
3. When the INIT page appears, check the position and time and press LSK 6 `←ACCEPT` on each unit. FPL MENU opens.
4. Type `LMML` and press LSK 1 (ORIGIN). Wait for the message `ORIGIN LMML`.
5. Type `LICC` and press RSK 1 (DEST).
6. Press LSK 3 `←DEP/ARR SID/STAR`, then `←DEP`. Select the departure runway and the SID, then press `ENTER`.
7. Press `FPL`. Add the en-route waypoints: type each name and press the LSK of the line where it goes, then `ENTER`. Close any discontinuity (select it, `DEL→`, `CONFIRM→`).
8. Press `VNAV`, then `NEXT` until VNAV CLB: type the cruise altitude, press the LSK of CRZ ALT and `ENTER`.
9. Check the route on the Kontur displays and the legs on the FPL page.

## Climb

After take-off, follow the SID. Use the FPL page to watch the active (magenta) waypoint, and the NAV page for the cross-track error and the time to the next waypoint. When cleared, the ABSU can follow the UNS-1 route in its navigation (НВУ) mode with the UNS–CHC switch at UNS.

## Cruise

If ATC clears you direct to a waypoint, press `DTO` and the LSK of that waypoint, or select it on the FPL page and press RSK 2 `DTO→`.

## Descent preparations

Before the top of descent, press `MENU`, LSK 3 `←DEP/ARR SID/STAR`, then `ARR→`. Select the approach, the STAR and its transition, press `ENTER`, and check the arrival on the FPL page. Remove any discontinuity between the en-route part and the STAR.

## Descent

On the PATH VNAV page (page 1) enter the planned descent rate as TGT V/S: TO TOD shows the distance to the top of descent, and after it the vertical speed on the FR PPOS line tells you how to stay on the path to the next altitude. The altitude constraints of the STAR are on the FPL page.

## Approach

Tune the ILS on the Tu-154M radio panels (the TUNE page shows what is set), and fly the approach with the aircraft's approach systems. If ATC holds you, use `HOLD` (see Holding).

## Shutdown

On each unit press `ON/OFF DIM` and `OFF→`, or switch off the UNS power. When no unit is on any more the flight plan is erased, ready for the next flight.

# CREDITS

UNS-1 for the Tu-154M v3 mod for X-Plane 12: screens, keys, 2D panel and this manual - the Tu-154M v3 project.

# COPYRIGHT

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